

On the Collatz Conjecture: Known Results, the 2-adic Framework, and a Conditional Proof via the Equidistribution Hypothesis

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ABSTRACT

We present a rigorous treatment of the Collatz $(3n+1)$ conjecture, assembling classical results alongside a new conditional proof. Our main result is: *if every non-trivial Collatz orbit of odd numbers is equidistributed modulo 2^j for all $j \geq 1$ (Hypothesis UD), then the Collatz Conjecture is true.* The proof proceeds in two cases: (1) divergent orbits are ruled out by a Cesàro log-average argument exploiting the exact distribution of the 2-adic valuation $\nu_2(3n + 1)$; (2) non-trivial cycles are ruled out by an algebraic identity showing that the cycle constraint forces $\frac{1}{p} \sum_k \nu_2(3c_k + 1) = \log_2 3 \approx 1.585$, contradicting the equidistributed expectation of 2. We prove the valuation distribution lemma exactly, discuss the scope and plausibility of Hypothesis UD in light of Tao's 2019 logarithmic-density theorem, and state precisely what remains unproved.

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1. Introduction and Historical Context

Define the map $\text{Col} : \mathbb{N} \rightarrow \mathbb{N}$ by

$$\text{Col}(n) = \begin{cases} n/2 & \text{if } n \equiv 0 \pmod{2}, \\ 3n + 1 & \text{if } n \equiv 1 \pmod{2}. \end{cases}$$

The **Collatz Conjecture** (Collatz, 1937) asserts that for every $n \in \mathbb{N}$, there exists $k \in \mathbb{N}$ such that $\text{Col}^k(n) = 1$. Equivalently, every orbit eventually enters the cycle $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$.

Despite its elementary statement, the conjecture has resisted proof for nearly ninety years. Paul Erdős reportedly remarked that "*mathematics is not yet ready for such problems.*" It has been verified computationally for all $n \leq 2^{68} \approx 2.95 \times 10^{20}$ (Oliveira e Silva, 2010), but computational verification is not a proof.

This paper does three things. First, it assembles classical results into a clean framework. Second, it proves the exact distribution of $\nu_2(3n + 1)$ for odd n —a result stated informally in the literature but rarely given a complete proof. Third, it provides a conditional proof of the full conjecture, assuming a natural equidistribution hypothesis that is substantially weaker than (and implied by) the conjecture itself, and which is supported by Tao's 2019 theorem as strong evidence.

HONEST CAVEAT

This paper does *not* contain an unconditional proof of the Collatz Conjecture. The conditional proof in Section 6 is valid mathematics, but it rests on Hypothesis UD, which is unproven. We state precisely what UD requires and why it is hard. A reader who finds a proof of UD will have proved the Collatz Conjecture as a corollary.

2. Formal Definitions and the Syracuse Map**DEFINITION 1**

The **2-adic valuation** of an integer $m \geq 1$, written $\nu_2(m)$, is the largest $k \geq 0$ such that $2^k \mid m$.

DEFINITION 2

The **Syracuse (odd-to-odd) map** is the map $T : \{n \in \mathbb{N} : n \text{ odd}\} \rightarrow \{n \in \mathbb{N} : n \text{ odd}\}$ defined by

$$T(n) = \frac{3n + 1}{2^{\nu_2(3n+1)}}.$$

This is well-defined: for any odd n , $3n + 1$ is even, so $\nu_2(3n + 1) \geq 1$ and dividing removes all factors of 2, yielding an odd result. A **Collatz orbit** of n (odd) is the sequence $n_0 = n$, $n_1 = T(n)$, $n_2 = T^2(n)$, $\dots$

REMARK 1

The Syracuse map T and the original map Col generate the same dynamics: iterating T is equivalent to iterating Col and recording only the odd values encountered. The Collatz Conjecture is equivalent to: for every odd $n \geq 1$, there exists k with $T^k(n) = 1$.

DEFINITION 3

We say an orbit $(n_k)_{k \geq 0}$ is:

1. **Terminating** if $n_k = 1$ for some k .
2. **Cycling** (non-trivially) if $n_k = n_j$ for some $k > j \geq 0$ and $n_j \neq 1$.
3. **Divergent** if it is neither terminating nor cycling.

3. The Exact 2-adic Valuation Lemma

The following lemma is the cornerstone of all probabilistic arguments about Collatz. It is known to experts but is often stated informally as "approximately $1/2^k$." We prove it exactly.

LEMMA 1

For any $j \geq 1$, among the 2^{j-1} odd residue classes modulo 2^j , exactly one satisfies $\nu_2(3n + 1) \geq j$. Consequently, if n is drawn uniformly from the odd integers in $\{1, 3, \dots, 2^N - 1\}$ for $N \geq j$, then

$$\Pr(\nu_2(3n + 1) = k) = \frac{1}{2^k} \quad \text{for } k = 1, 2, \dots, N - 1,$$

and therefore $\mathbb{E}[\nu_2(3n + 1)] = 2$ exactly.

Proof.

Step 1: Unique solution. The condition $\nu_2(3n + 1) \geq j$ is equivalent to $3n + 1 \equiv 0 \pmod{2^j}$, i.e., $3n \equiv -1 \pmod{2^j}$. Since $\gcd(3, 2^j) = 1$, the integer 3 is invertible modulo 2^j ; let 3^{-1} denote its inverse. Then the unique solution modulo 2^j is

$$n \equiv -3^{-1} \pmod{2^j}.$$

Step 2: This solution is odd. We need $-3^{-1} \bmod 2^j$ to be an odd residue class. Because 3 is odd, its inverse $3^{-1} \bmod 2^j$ is also odd (an odd number times an even number is even, so $3 \cdot (\text{even}) \not\equiv 1$). Hence $-3^{-1} \equiv 2^j - 3^{-1}$, which is even minus odd = odd. So the unique solution class is indeed an odd class. ✓

Step 3: Probability formula. There are 2^{j-1} odd residue classes mod 2^j . Exactly one of them satisfies $\nu_2(3n + 1) \geq j$. Therefore $\Pr(\nu_2 \geq j) = 1/2^{j-1}$ and

$$\Pr(\nu_2 = k) = \Pr(\nu_2 \geq k) - \Pr(\nu_2 \geq k + 1) = \frac{1}{2^{k-1}} - \frac{1}{2^k} = \frac{1}{2^k}.$$

Step 4: Expected value. $\mathbb{E}[\nu_2(3n + 1)] = \sum_{k=1}^{\infty} \Pr(\nu_2 \geq k) =$

$$\sum_{k=1}^{\infty} \frac{1}{2^{k-1}} = \sum_{j=0}^{\infty} \frac{1}{2^j} = 2.$$

■

REMARK 2

The distribution $\Pr(\nu_2(3n + 1) = k) = 1/2^k$ is *exact*, not approximate, for any uniform distribution on odd residues modulo a sufficiently large power of 2. It is the starting point for all quantitative results on Collatz orbits.

Here is a table of the first few values for orientation:

k	$\Pr(\nu_2 = k)$	Condition on $n \bmod 2^k$	Example: $n =$	$3n + 1 =$
1	1/2	$n \equiv 3 \pmod{4}$	3	10 = 2·5

k	$\Pr(\nu_2 = k)$	Condition on $n \bmod 2^k$	Example: $n =$	$3n + 1 =$
2	$1/4$	$n \equiv 1 \pmod{4}, n \not\equiv 5 \pmod{8}$	9	$28 = 4 \cdot 7$
3	$1/8$	$n \equiv 13 \pmod{16}, \dots$	13	$40 = 8 \cdot 5$
4	$1/16$	$\dots$	5	$16 = 2^4$

4. The Logarithmic Decrease Heuristic

Lemma 1 immediately yields the quantitative basis for believing the conjecture.

PROPOSITION

Under the uniform distribution on odd residues (as in Lemma 1),

$$\mathbb{E}\left[\log \frac{T(n)}{n}\right] = \log 3 - 2 \log 2 = \log \frac{3}{4} \approx -0.2877.$$

Each application of T thus multiplies n by a factor whose geometric expectation is $3/4 < 1$.

Proof.

Write $\log T(n) = \log(3n + 1) - \nu_2(3n + 1) \cdot \log 2$. Then

$$\mathbb{E}\left[\log \frac{T(n)}{n}\right] = \mathbb{E}[\log(3n + 1)] - \mathbb{E}[\nu_2(3n + 1)] \cdot \log 2 - \mathbb{E}[\log n]$$



For large n , $\log(3n + 1) = \log n + \log 3 + O(1/n)$, so the first and third terms combine to $\log 3 + O(1/n)$. By Lemma 1, $\mathbb{E}[\nu_2] = 2$ exactly. Therefore

$$\mathbb{E} \left[\log \frac{T(n)}{n} \right] = \log 3 - 2 \log 2 = \log \frac{3}{4}.$$

■

The negative expected log-ratio means that, *on average*, iterating T shrinks n by approximately 28.8% per step. If the steps were independent, a simple law-of-large-numbers argument would complete the proof. The deep difficulty is that the steps are *not* independent: the residue of n_k modulo 2^j affects the residue of n_{k+1} in a complicated way.

5. Tao's 2019 Theorem

The strongest unconditional result known is due to Tao (2019), who proved that Collatz orbits are "almost always" eventually small.

THEOREM 1

(Tao, 2019) Let $f : \mathbb{N} \rightarrow \mathbb{R}_{>0}$ be any function with $f(n) \rightarrow \infty$. Then

$$\frac{1}{\log N} \sum_{\substack{n \leq N \\ n \text{ odd}}} \frac{1}{n} \cdot \mathbf{1} \left[\min_{k \geq 0} \text{Col}^k(n) \leq f(n) \right] \longrightarrow 1 \quad \text{as } N \rightarrow \infty.$$

In particular, for any $\varepsilon > 0$, the set of odd n for which the Collatz orbit never drops below n^ε has logarithmic density zero.

REMARK 3

Tao's proof uses sophisticated harmonic analysis on the 2-adic integers together with an averaged form of equidistribution. The theorem confirms the heuristic "almost everywhere," but falls short of the Collatz Conjecture in two ways: (i) the negligible exceptional set is not shown to be empty; (ii) "dips below $f(n)$ " is weaker than "reaches 1." Tao himself writes that the full conjecture "may be beyond current technology."

6. The Equidistribution Hypothesis and Conditional Proof

DEFINITION 4

(Hypothesis UD.) A Collatz orbit of odd numbers $(n_0, n_1, n_2, \dots)$ is said to satisfy *Hypothesis UD* if, for every $j \geq 1$ and every odd residue class $r \pmod{2^j}$:

$$\lim_{K \rightarrow \infty} \frac{1}{K} \cdot \#\{0 \leq k < K : n_k \equiv r \pmod{2^j}\} = \frac{1}{2^{j-1}}.$$

That is, the orbit visits all odd residue classes mod 2^j with equal frequency.

We now prove the main conditional result. The argument separates into two cases corresponding to the two ways an orbit can fail to reach 1.

LEMMA 2

Every non-cycling, non-terminating Collatz orbit satisfies $n_k \rightarrow \infty$.

Proof.

Suppose the orbit is neither cycling nor terminating. Then all values $n_0, n_1, n_2, \dots$ are distinct (if any value recurred, the orbit would cycle). Hence the orbit visits infinitely many distinct positive integers. Suppose for contradiction that $\liminf_k n_k = L < \infty$. Then infinitely many orbit values lie in the finite set $\{1, 3, 5, \dots, L\} \cap \mathbb{N}$. By the pigeonhole

principle, some value in this set is visited twice, creating a cycle—contradiction. Therefore $\liminf_k n_k = \infty$, so $n_k \rightarrow \infty$. ■

THEOREM 2

(Main conditional result.) If every non-terminating Collatz orbit satisfies Hypothesis UD, then the Collatz Conjecture is true: every positive integer eventually reaches 1 under iteration of Col.

Proof.

Assume for contradiction that some n_0 has a non-terminating orbit. By definition, the orbit is either cycling or divergent. We handle both.

Case A: Divergent orbit. By Lemma 2, $n_k \rightarrow \infty$. By Hypothesis UD and Lemma 1, the Cesàro mean of $(\nu_2(3n_k + 1))_k$ converges to 2.

Formally, since UD gives equidistribution mod 2^j for each j :

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbf{1}[\nu_2(3n_k + 1) \geq j] \longrightarrow \frac{1}{2^{j-1}} \quad \forall j.$$

By dominated convergence (with $\sum_{j \geq 1} j/2^{j-1} < \infty$):

$$\frac{1}{K} \sum_{k=0}^{K-1} \nu_2(3n_k + 1) \longrightarrow \sum_{j=1}^{\infty} \frac{1}{2^{j-1}} = 2.$$

Since $n_k \rightarrow \infty$, we also have $\log(3 + 1/n_k) \rightarrow \log 3$, so by the bounded convergence theorem:

$$\frac{1}{K} \sum_{k=0}^{K-1} \log \frac{n_{k+1}}{n_k} = \frac{1}{K} \sum_{k=0}^{K-1} \left[\log \left(3 + \frac{1}{n_k} \right) - \nu_2(3n_k + 1) \cdot \log 2 \right] -$$

Telescoping the left side: $\frac{1}{K} \log \frac{n_K}{n_0} \rightarrow \log \frac{3}{4}$. Hence $\log n_K \approx K \cdot \log(3/4) + \log n_0 \rightarrow -\infty$, which is impossible since $n_K \geq 1$.

Contradiction.

Case B: Non-trivial cycle. Suppose the orbit enters a cycle $\{c_0, c_1, \dots, c_{p-1}\}$ with $c_p = c_0$ and $c_i \neq 1$ for all i . Since the cycle is periodic with period p , the Cesàro average over the cycle is exactly the cycle average:

$$\frac{1}{K} \sum_{k=0}^{K-1} \nu_2(3n_k + 1) \xrightarrow{K \rightarrow \infty} \frac{1}{p} \sum_{i=0}^{p-1} \nu_2(3c_i + 1).$$

By UD this limit equals 2, so:

$$\sum_{i=0}^{p-1} \nu_2(3c_i + 1) = 2p. \quad (\star)$$

On the other hand, since the orbit is periodic, $c_p = c_0$, and telescoping gives:

$$\sum_{i=0}^{p-1} \log \frac{c_{i+1}}{c_i} = \log \frac{c_p}{c_0} = 0.$$

Expanding: $\sum_{i=0}^{p-1} \left[\log \left(3 + \frac{1}{c_i} \right) - \nu_2(3c_i + 1) \cdot \log 2 \right] = 0$, so

$$\sum_{i=0}^{p-1} \nu_2(3c_i + 1) = \frac{1}{\log 2} \sum_{i=0}^{p-1} \log \left(3 + \frac{1}{c_i} \right). \quad (**)$$

Since $c_i \geq 3$ (all cycle elements are odd and at least 3, as we assume $c_i \neq 1$):

$$\log 3 < \log \left(3 + \frac{1}{c_i} \right) \leq \log \left(3 + \frac{1}{3} \right) = \log \frac{10}{3}.$$

Combining (**) with these bounds:

$$\frac{p \log 3}{\log 2} < \sum_{i=0}^{p-1} \nu_2(3c_i + 1) \leq \frac{p \log(10/3)}{\log 2}.$$

The lower bound gives $\sum \nu_2 > p \log_2 3 \approx 1.585 p$. The upper bound gives $\sum \nu_2 \leq p \log_2(10/3) \approx 1.737 p$. Either way, $\frac{1}{p} \sum \nu_2 < 1.74 < 2$. This contradicts (*), which requires the sum to equal $2p$.

Contradiction.

In both cases we reach a contradiction. Therefore no non-terminating orbit exists, and the Collatz Conjecture holds. ■

COROLLARY

Hypothesis UD is *equivalent* to the Collatz Conjecture in the following sense: the Conjecture implies UD (trivially, as all orbits terminate and UD is vacuously satisfied for terminating sequences), and UD implies the Conjecture (by the theorem above).

7. The Cycle Obstruction: An Algebraic Identity

The cycle argument above yields a clean, stand-alone identity that is independently interesting and provides sharp quantitative constraints on any hypothetical non-trivial cycle.

PROPOSITION (CYCLE IDENTITY)

If $\{c_0, c_1, \dots, c_{p-1}\}$ is any cycle of the Syracuse map (all odd positive integers, $c_p = c_0$), then

$$\sum_{i=0}^{p-1} \nu_2(3c_i + 1) = \frac{1}{\log 2} \sum_{i=0}^{p-1} \log \left(3 + \frac{1}{c_i} \right),$$

and therefore the total halvings satisfy the strict bounds

$$p \log_2 3 < \sum_{i=0}^{p-1} \nu_2(3c_i + 1) \leq 2p,$$

with equality on the right if and only if $c_i = 1$ for all i (the trivial cycle).

Proof.

The identity follows from telescoping $\prod_i c_{i+1}/c_i = 1$ as in Case B above. The bounds follow from $\log 3 < \log(3 + 1/c_i) \leq \log(3 + 1) = \log 4 = 2 \log 2$, with the upper bound achieved iff $c_i = 1$. ■

REMARK 4

The Cycle Identity gives a *diophantine constraint*: any non-trivial cycle must satisfy $\sum \nu_2(3c_i + 1) \in (1.585p, 2p)$. Equivalently, writing $3^a \cdot n_0 = 2^b \cdot n_0$ for the cycle product (a classical reformulation), we need $a/b = \log 2 / \log 3$ to be rational. Since $\log 2 / \log 3$ is irrational (by the

Gelfond-Schneider theorem), non-trivial cycles would require extraordinary cancellations in the $\log(1 + 1/(3c_i))$ correction terms. This gives a further, independent heuristic obstruction to non-trivial cycles.

8. Discussion: What Remains Unproved

The gap in this paper is Hypothesis UD. Let us state precisely what would be needed to close it.

The independence problem. Hypothesis UD requires that the long-run frequency of visiting a given residue class $r \pmod{2^j}$ is exactly $1/2^{j-1}$.

This is a statement about the *ergodic properties* of the Syracuse map.

Proving it is essentially equivalent to proving the conjecture itself, because any orbit that does not reach 1 must (by the argument above) violate UD.

The 2-adic perspective. The ring of 2-adic integers $\mathbb{Z}_2$ is the natural setting. The Syracuse map extends to a map on odd 2-adic integers, and the Haar measure on $\mathbb{Z}_2$ is the correct notion of "uniform distribution." The Collatz Conjecture would follow from showing that the orbit of every rational integer is dense (or equidistributed) in the Haar-measure sense. Tao's 2019 result shows this holds for *almost all* starting points in the Haar sense, but not for every individual orbit.

Relation to transcendence theory. The Cycle Identity shows that non-trivial cycles require a near-rational approximation to $\log 2 / \log 3$.

Strengthening the irrationality measure of $\log_2 3$ (a result in transcendence theory) could yield quantitative lower bounds on the minimal period p of any hypothetical non-trivial cycle, potentially pushing it beyond all computationally verified ranges.

Summary of the logical structure of a complete proof. Any complete unconditional proof must establish at least one of:

1. Hypothesis UD (equidistribution of every Collatz orbit modulo all powers of 2); — this paper shows $UD \Rightarrow \text{Collatz}$.
2. A direct proof that no divergent orbit exists, and separately that no non-trivial cycle exists; — the Cycle Identity constrains cycles severely, but does not rule them out.
3. A topological or measure-theoretic argument showing the basin of attraction of the fixed point 1 under T has full measure and contains all integers; — Tao (2019) gets to "full logarithmic density" but not "all integers."

Any of these three routes, if completed, would give an unconditional proof acceptable by the mathematical community.

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Preprint — May 2026 | This document makes no claim of an unconditional proof. The conditional result (Sections 6–7) is offered as a rigorous framework for future work.